

Tien-Chun (Leo) Chen

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SUMMARY

Robotics Controls & Sim-to-Real Engineer with 2 years of industry experience in humanoid locomotion and autonomous vehicles. Experienced in residual RL, real-time control, system identification, and hardware debugging.

WORK EXPERIENCE

FIH Co., Ltd. (Subsidiary of Foxconn Group), New Taipei, Taiwan 03/2025 – 06/2026

Robotics Research Engineer – Humanoid Locomotion, Software Research Center

- Diagnosed a pure-RL gait that walked in Isaac Sim but failed in MuJoCo and on the proprietary humanoid, tracing the sim-to-real gap to actuator latency, bandwidth limits, and unmodeled parallel-linkage dynamics.
- Achieved 30 minutes of continuous walking without operator intervention or controller resets on the proprietary humanoid with linear-actuator-driven parallel-linkage legs by combining LIPM trajectory priors with residual RL.
- Built an Isaac Lab residual-RL pipeline with reward shaping and domain randomization over actuator dynamics and latency, aligning policy training with measured bandwidth constraints for sim-to-real deployment.
- Developed CasADi-generated C++ virtual-work mappings for 8 linear-actuator parallel-linkage joints, using Jacobian-determinant and actuator-stroke analysis to define singularity-free ranges for 1 kHz control.
- Implemented impedance and joint-level force control for the parallel-linkage leg joints, compensating actuator nonlinearities and validating closed-loop bandwidth through chirp-sweep testing.

Kingwaytek Technology Co., Ltd., Taoyuan, Taiwan

05/2024 – 02/2025

Autonomous Vehicle R&D Software Engineer – Planning, Software Research Center

- Built a real-time ROS2 System/Topic Health Monitor spanning sensing, localization, perception, and planning, enabling field engineers to distinguish upstream data degradation from true planning infeasibility.
- Applied trajectory optimization tuned against intervention logs, balancing kinematic feasibility, collision margins, and ride comfort to raise autonomous operation rate from 75% to 93% in constrained industrial environments.
- Redesigned intersection navigation by representing junctions as geometric polygons beyond HD-map lane boundaries, enabling constrained trajectory generation and dynamic-obstacle avoidance in confined spaces.
- Implemented Autoware-based fleet dispatch and reverse-docking workflows, enabling automated alignment with wireless-charging infrastructure.

TECHNICAL PROJECTS

Embodied AI Ping-Pong Capstone, UC Berkeley 08/2026 – present

- Developing whole-body control for locomotion and manipulation on the AgiBot A3 humanoid, extending Berkeley's HITTER baseline toward generalized target selection and strike planning; implementation and validation in progress.

Vision-Based Manipulation and Grasping, The Construct Final Project

02/2024 – 04/2024

- Built a MoveIt 2 pick-and-place pipeline using depth-camera point clouds, least-squares circle fitting, constrained Cartesian motion, and dynamic planning-scene updates, eliminating false collision aborts during execution.

Autonomous Mobile Robot Research, NYCU Robotics Lab

11/2021 – 08/2022

- Built a differential-drive AMR from mechanical and electrical integration through dual-LiDAR calibration, low-level motor control, Cartographer SLAM, and autonomous navigation, validating the complete stack on physical hardware.

EDUCATION

University of California, Berkeley, Berkeley, CA, USA

Expected 05/2027

M.Eng. in Electrical Engineering and Computer Sciences (EECS) – Concentration: Robotics and Embedded Software

- Graduate coursework: Convex Optimization, Optimal Control, and Model Predictive Control

National Yang Ming Chiao Tung University (NYCU), Hsinchu, Taiwan

09/2018 – 07/2022

Bachelor of Science in Mechanical Engineering; Double Major in Electrical Engineering

- Overall GPA: 3.85/4.30, Major GPA: 4.11/4.30 | Top 3% Academic Excellence Awards, 2021 & 2022

TECHNICAL SKILLS

- Controls & Optimization: Impedance and force control, system identification, trajectory optimization, real-time control
- Robot Learning & Simulation: Residual RL, PPO, PyTorch, LibTorch, Isaac Lab/Gym, Isaac Sim, MuJoCo, Gazebo
- Robotics Software: C/C++, Python, ROS2, Autoware, MoveIt 2, Nav2, PCL, CasADi, Pinocchio, Linux, Git, Docker